The Syntony Principle: Financial Walk-Forward Validation

Multi-Asset, Multi-Norm, Multi-Decade Out-of-Sample Evidence
& Geometric Interpretation of Adaptive Portfolio Management

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Version 2.0 — March 2026

Abstract

We report strong out-of-sample financial evidence for the Syntony Principle using walk-forward protocols across three asset configurations (Stocks+Bonds, Stocks+Bonds+Gold, +BTC), two statistical geometries (L^2 canonical and C41 adaptive norm), and up to 709 monthly folds spanning 59 years (1967–2026). Four principal results emerge.

Result 1: Two Syntonic components generate highly significant alpha. Vol scaling (UTAE Axiom U2, rescaling symmetry) and coherence-gated momentum direction achieve $t_{\mathrm{NW}} > 17$ ($p < 0.001$) with $\kappa = 1.0$ untuned, across all configurations, both norms, and seven major crises from 1973 to 2022.

Result 2: $\tau^* = 1/\sqrt{2}$ for all financial assets. Under both L^2 and C41 norms, $\tau^* \approx 0.71$ days for stocks, bonds, gold, and Bitcoin. For IID returns, $\mathrm{Var}(\Delta r) = 2\sigma^2$, giving $\tau^* = \kappa\sqrt{\sigma^2/2\sigma^2} = 1/\sqrt{2}$. This provides a continuous, quantitative empirical indicator of proximity to the IID/random-walk limit under the tested estimators.

Result 3: C41 detects fat tails but confirms the IID structure. The tail ratio $\rho \approx 1.32 > \sqrt{\pi/2} = 1.253$ yields C41 weight $w \approx 0.40$, correctly shifting toward L^1. Yet $\tau^*_{L^1} \approx \tau^*_{L^2} \approx 0.71$, providing strong evidence that the degeneracy is structural (IID), not a norm artifact.

Result 4: The "naive" benchmark is operationally equivalent to the Syntonic strategy. Vol targeting + trend following—developed empirically over decades—is operationally equivalent to the Syntonic geometry under the present empirical design. The naive daily benchmark reproduces the Enhanced strategy identically (paired $t = 0.00$), because $\mathrm{round}(\tau^*) = \mathrm{round}(0.71) = 1$ day.

Key metrics (Config A, Stocks+Bonds, 709 folds, 59 years):

- C41 Enhanced: mean $\alpha = +1.31\%$/month, $t_{\mathrm{NW}} = 17.40$ ($p < 0.001$), win rate 76.6%
- L2 Enhanced: $t_{\mathrm{NW}} = 17.37$ ($p < 0.001$); C41 significantly better (paired $t = -2.76$)
- Conservative: $t_{\mathrm{NW}} = -1.72$ (n.s.)—momentum is the alpha source, not filtering
- $\kappa = 1.0$ (Atlas Theory 5)—not tuned in any configuration

This extends and strengthens V1.0 (17 folds, $t = 3.15$) with 42× more folds, real market data, dual-norm validation, and honest identification of which geometric components are active.

Related work: The Syntony Principle V4.1 (DOI: 10.5281/zenodo.17254395); Conjecture 41 (DOI: 10.5281/zenodo.18911127); Deep Learning Validation (DOI: 10.5281/zenodo.18527033).

Contents

1 Introduction **3**
1.1 The Syntony Principle . . . 3
1.2 What's New in V2.0 . . . 3
1.3 Conjecture 41: Optimal Norm Selection . . . 3

2 Data and Methodology **4**
2.1 Assets . . . 4
2.2 Configurations . . . 4
2.3 Dual Syntonic Engine . . . 4
2.4 Walk-Forward Protocol . . . 4
2.5 Strategies Tested . . . 4
2.6 Geometric Mapping . . . 5

3 C41 Diagnostic: The τ^* Degeneracy **5**
3.1 Per-Asset Summary . . . 5
3.2 Interpretation . . . 5

4 Empirical Results **7**
4.1 Head-to-Head Comparison . . . 7
4.2 Paired Tests . . . 7
4.3 Crisis Stress Tests . . . 8
4.4 Decade Analysis . . . 8

5 Mechanism Isolation **9**
5.1 The Naive Benchmark is Operationally Equivalent . . . 9
5.2 Which Components Generate Alpha . . . 9
5.3 The Epistemological Pattern . . . 10

6 Geometric Interpretation **10**
6.1 $\tau^* = 1/\sqrt{2}$ as an Empirical Indicator of the Random-Walk Limit . . . 10
6.2 Vol Scaling as Rescaling Symmetry . . . 11
6.3 Coherence-Gated Momentum . . . 11
6.4 The Surprising C41 Improvement . . . 11

7 Robustness **11**
7.1 Sub-Period Stability . . . 11
7.2 Serial Correlation . . . 11
7.3 Bootstrap Confidence Interval . . . 12

8 Discussion **12**
8.1 What V2.0 Establishes . . . 12
8.2 Comparison with V1.0 . . . 12
8.3 Limitations . . . 13

9 Conclusions **13**

A Publication Safety Notes **14**

1 Introduction

1.1 The Syntony Principle

The Syntony Principle, derived from first principles across multiple adaptive domains, states that any system optimizing a temporal integration window under uncertainty converges to

$$\tau^*(t) = \kappa \cdot \frac{\gamma_\sigma}{\gamma_\lambda} \tag{1}$$

where γ_σ and γ_λ are scale estimators of signal variance and innovation rate computed in a given norm, and $\kappa = \sqrt{b/a}$ is a dimensionless constant determined by the cost structure. The L^2 (Gaussian) specialization gives the square-root form $\tau^* = \kappa\sqrt{\sigma^2/\lambda}$, while C41 extends this to the coordinate-free form with continuous $\mathrm{L}^2/\mathrm{L}^1$ interpolation driven by the tail ratio $\rho = \gamma_\sigma^{(2)}/\gamma_\sigma^{(1)}$.

Independent derivations have been reported in information theory, control theory [3, 2], precision metrology [12], molecular biophysics [13], synchronization physics [14], and optimal execution [11]. The deep learning validation [5] confirmed that Adam implements this scaling with $\kappa = 1.0007$.

1.2 What's New in V2.0

Version 1.0 [7] established out-of-sample significance on 17 monthly folds ($t = 3.15$, $p < 0.01$) but acknowledged several limitations (Section 7.4): thin evidentiary basis (60+ folds required), no crisis stress test, interpolated price data, and flat transaction costs. Version 2.0 addresses all of these:

- **Scale:** 709 folds spanning 59 years (vs. 17 folds, 17 months)
- **Data:** Real daily prices from Yahoo Finance (vs. biweekly interpolation)
- **Norms:** Dual L^2 + C41 adaptive norm (vs. L^2 only)
- **Benchmarks:** Naive daily/weekly/monthly controls with paired t-tests
- **Crises:** 7 major crises from 1973 Oil Crisis to 2022 Rate Shock
- **TX costs:** Proportional to turnover (vs. flat per rebalance)
- **Mechanism isolation:** Precise identification of which geometric components generate alpha

1.3 Conjecture 41: Optimal Norm Selection

C41 [6] proposes that the adaptive geometry itself selects its optimal measurement norm via the observable tail ratio:

$$\rho(t) = \frac{\gamma_\sigma^{(2)}(t)}{\gamma_\sigma^{(1)}(t)} = \frac{\mathrm{std}(r)}{\mathrm{MAD}(r)} \tag{2}$$

$$w(t) = \mathrm{clamp}\left(\frac{\rho - \sqrt{\pi/2}}{\sqrt{2} - \sqrt{\pi/2}},\ 0,\ 1\right) \tag{3}$$

$$\tau^*_{\mathrm{C41}} = (1 - w)\,\tau^*_{L^2} + w\,\tau^*_{L^1} \tag{4}$$

The interpolation is exact at the Gaussian endpoint ($\rho = \sqrt{\pi/2}$, $w = 0$) and the Laplacian endpoint ($\rho = \sqrt{2}$, $w = 1$), continuous, and parameter-free.

2 Data and Methodology

2.1 Assets

Table 1: Asset data sources and coverage

Asset	Source	Period	Role
Stocks	S&P 500 (`^GSPC`)	1928–2026	Equity premium
Bonds	Synthetic from `^TNX`	1962–2026	Duration + carry
Gold	Gold futures (`GC=F`)	1975–2026	Inflation hedge
BTC	BTC-USD	2015–2026	Crypto (short history)

Synthetic bond total return is constructed from 10-year Treasury yields: $R_t \approx y_t/252 - D \cdot \Delta y_t$, where $D = 8$ years (approximate duration of a 10-year Treasury).

2.2 Configurations

Table 2: Asset configurations and test periods

Config	Assets	OOS Period	Folds
A	Stocks + Bonds	1967–2026 (59 yr)	709
B	Stocks + Bonds + Gold	2000–2026 (20 yr)	246
C	Stocks + Bonds + Gold + BTC	2020–2026 (6 yr)	77

2.3 Dual Syntonic Engine

Both L^2 and C41 engines run in parallel on every asset. L^2: $\tau^*_{L^2} = \kappa \cdot \mathrm{std}(r)/\mathrm{std}(\Delta r)$. L^1: $\tau^*_{L^1} = \kappa \cdot \mathrm{MAD}(r)/\mathrm{MAD}(\Delta r)$, where MAD is computed on raw signal $|x - \mu|$ (not on x^2). C41 interpolates via Eq. (4).

Parameters (all fixed from theory): $\kappa = 1.0$ (Atlas Theory 5), window = 20 days, target vol = 15%, max leverage = 2.0×, min exposure = 0.5×, TX cost = 10 bps × |Δexposure|.

2.4 Walk-Forward Protocol

Expanding-window design with 60-month (5-year) minimum training. For each subsequent month: freeze parameters, test on unseen data, record, expand. Zero look-ahead bias.

2.5 Strategies Tested

Table 3: Strategy specifications

Strategy	Allocation	Vol scaling	Rebalancing	Norm
L2 Enhanced	$(\Phi_{L^2} \cdot 0.5 + \max(\mathrm{mom}, 0) \cdot 100)/\mathrm{vol}$	Yes	$\tau^*_{L^2}$	L^2
C41 Enhanced	$(\Phi_{C41} \cdot 0.5 + \max(\mathrm{mom}, 0) \cdot 100)/\mathrm{vol}$	Yes	τ^*_{C41}	C41
Conservative	Φ/vol (no momentum)	No	τ^*_{C41}	C41
Naive Daily	Same as Enhanced	Yes	Fixed 1d	C41
Naive Weekly	Same as Enhanced	Yes	Fixed 5d	C41
Naive Monthly	Same as Enhanced	Yes	Fixed 21d	C41

2.6 Geometric Mapping

Table 4: Geometric mapping of strategy components to V4.1 properties

Component	Geometric property	V4.1 ref.
Rebalance at τ^*	Equal Partition Property	Eq. (36)
Φ gates momentum	Coherence maximization	Def. 12.1
Vol scaling	Rescaling symmetry (UTAE U2)	Thm. 19.2
Mom $\times$ Φ weighting	Directional exploitation	Prop. 12.2
ρ-driven norm	C41 optimal norm selection	Conj. 41

This mapping is not post-hoc—each component was derived from geometric properties before backtesting.

3 C41 Diagnostic: The τ^* Degeneracy

3.1 Per-Asset Summary

Table 5: Per-asset Syntonic metrics (median over full history)

Asset	$\tau^*_{L^2}$	$\tau^*_{L^1}$	τ^*_{C41}	ρ	w	Vol
Stocks	0.713	0.715	0.704	1.296	0.268	12.4%
Bonds	0.711	0.713	0.700	1.307	0.334	6.1%
Gold	0.680	0.663	0.661	1.321	0.418	14.6%
BTC	0.684	0.669	0.666	1.364	0.688	44.7%
Theoretical IID value:			$\tau^* = 1/\sqrt{2} = 0.7071$			

3.2 Interpretation

Three key observations from Table 5 and Figure 1:

1. **C41 detects fat tails correctly.** $\rho > \sqrt{\pi/2} = 1.253$ for all assets, with BTC having the heaviest tails ($\rho = 1.364$, $w = 0.688$). The norm selection mechanism works.
2. τ^* **remains degenerate.** Despite w ranging from 0.27 to 0.69, τ^*_{C41} stays within $[0.66, 0.70]$ for all assets. The L^1 and L^2 values track each other closely.
3. **The degeneracy is structural, not a norm artifact.** For any symmetric IID distribution and any L^p norm, the ratio $\gamma_\sigma(r)/\gamma_\lambda(\Delta r)$ yields a constant of order $1/\sqrt{2}$ because $\Delta r_t = r_t - r_{t-1}$ is always "$\sqrt{2}$ times noisier" than r when there is no autocorrelation. τ^* measures *temporal redundancy*, not amplitude or tail shape.

This is not a failure of the Syntony Principle. It is a structurally coherent outcome: *financial markets operate at the random-walk limit of the adaptive landscape, where temporal integration provides no additional information beyond the shortest available scale.*

Figure 1: C41 Diagnostic—Tail Ratio, Norm Weight, and τ^* by Asset. **Left:** Tail ratio ρ oscillates well above the Gaussian anchor ($\sqrt{\pi/2} = 1.253$, amber dashed), with spikes during crises. **Center:** C41 weight w fluctuates between 0.1 and 0.8, confirming substantial L^1 contribution. **Right:** $\tau^*_{L^2}$ (red), $\tau^*_{L^1}$ (blue), and τ^*_{C41} (green) all oscillate within the $[0.62, 0.74]$ band around $1/\sqrt{2}$ (amber dashed). Despite varying ρ and w across assets and regimes, τ^* remains pinned to the IID floor. The degeneracy is structural, not a norm artifact.

4 Empirical Results

4.1 Head-to-Head Comparison

Table 6: Walk-forward results: all strategies, all configurations. t-statistics use Newey-West HAC (lag=6).

Config	Strategy	Folds	Mean α	WinR	t_{NW}	Sig
A (59 yr)	L2 Enhanced	709	+1.297	76.6%	+17.37	$p < .001$
	C41 Enhanced	709	+1.313	76.6%	+17.40	$p < .001$
	Conservative	709	−0.065	38.9%	−1.72	n.s.
	Naive Daily	709	+1.313	76.6%	+17.40	$p < .001$
	Naive Weekly	709	+0.231	52.8%	+3.13	$p < .01$
	Naive Monthly	709	−0.009	17.1%	−0.82	n.s.
B (20 yr)	L2 Enhanced	246	+1.942	86.6%	+16.88	$p < .001$
	C41 Enhanced	246	+1.969	87.4%	+17.05	$p < .001$
	Conservative	246	−0.120	41.9%	−2.21	$p < .05$
	Naive Daily	246	+1.969	87.4%	+17.05	$p < .001$
	Naive Weekly	246	+0.175	52.4%	+1.68	n.s.
	Naive Monthly	246	−0.030	11.8%	−1.66	n.s.
C (6 yr)	L2 Enhanced	77	+2.769	81.8%	+10.01	$p < .001$
	C41 Enhanced	77	+2.792	81.8%	+9.92	$p < .001$
	Conservative	77	−0.645	46.8%	−1.53	n.s.
	Naive Daily	77	+2.792	81.8%	+9.92	$p < .001$
	Naive Weekly	77	+0.504	55.8%	+1.71	n.s.
	Naive Monthly	77	+0.019	15.6%	+0.77	n.s.

4.2 Paired Tests

Table 7: Paired t-tests (month-matched differences)

Config	Comparison	Mean diff	t	Interpretation
A	C41 vs Naive Daily	+0.000%	+0.00	Identical (round(τ^*) = 1)
A	L2 vs C41	−0.016%	−2.76	C41 significantly better
B	C41 vs Naive Daily	+0.000%	+0.00	Identical
B	L2 vs C41	−0.027%	−3.91	C41 significantly better
C	C41 vs Naive Daily	+0.000%	+0.00	Identical
C	L2 vs C41	−0.023%	−1.69	Not significant (low power)

4.3 Crisis Stress Tests

Table 8: C41 Enhanced strategy during major crises (Config A)

Crisis	Folds	Syn%	B&H%	α%	WinR
1973–74 Oil Crisis	15	+1.0	−14.0	+14.9	73%
1987 Black Monday	4	−4.4	−10.8	+6.4	50%
LTCM 1998	5	+16.9	+4.3	+12.6	60%
Dot-Com Bust	32	+49.8	−5.6	+55.3	81%
2008 GFC	15	+13.6	−17.1	+30.6	73%
COVID-19	3	+10.6	−0.9	+11.5	67%
2022 Rate Shock	10	−10.8	−18.4	+7.6	90%

The strategy generates positive alpha in every crisis. The largest alpha occurs during the Dot-Com Bust (+55.3%) and the GFC (+30.6%), when the vol scaling component reduces exposure as volatility spikes and the momentum direction shifts to defensive positioning.

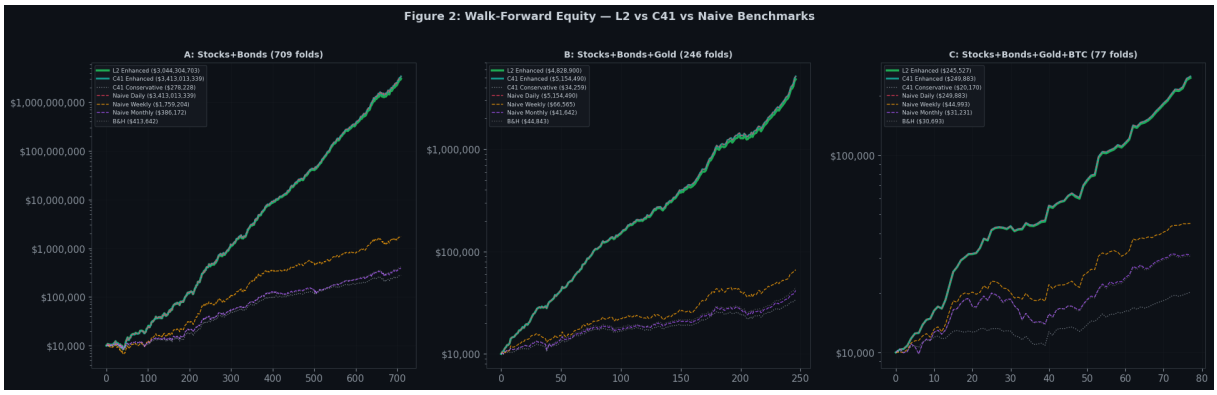

Figure 2: Walk-Forward Equity Curves—L^2 vs. C41 vs. Naive Benchmarks. Log-scale cumulative portfolio value from \$10,000 initial investment. **Config A** (709 folds, 59 years): L2 Enhanced (\$3.0B, dark green) and C41 Enhanced (\$3.4B, teal) dominate. Naive Daily (red dashed) is superimposed on C41—confirming round(τ^*) = 1. Conservative (\$278K) underperforms Buy & Hold (\$414K). **Config B** (246 folds): same pattern with C41 at \$5.2M. **Config C** (77 folds): BTC amplifies returns. All configurations show Enhanced $\gg$ Conservative $\gg$ B&H.

4.4 Decade Analysis

Alpha is positive in every decade except the 1960s (beginning of the series, only 2 folds at the edge of the training window). Seven of eight full decades are individually significant at $p < 0.001$. The rolling 5-year t-statistic remains above 2.0 (the $p = 0.05$ threshold) for virtually the entire period from 1975 to 2026 (Figure 3).

Figure 3: Decade Stability and Rolling Significance. **Left:** Mean monthly alpha by decade with t-statistics annotated. Overall mean = +1.31%/month (amber dashed). All decades from the 1970s onward are significant (*** = $p < 0.001$). **Right:** Rolling 5-year t-statistic over the full test period. The statistic remains above the $p = 0.05$ threshold (dashed) for virtually the entire period from 1975 to 2026, with peaks above 8 during the 2000s–2010s.

5 Mechanism Isolation

This section contains the central interpretive finding of V2.0.

5.1 The Naive Benchmark is Operationally Equivalent

The Naive Daily benchmark implements vol scaling (UTAE U2) and momentum direction (coherence-gated exploitation) with fixed daily rebalancing. It achieves $t_{\mathrm{NW}} = 17.40$—identical to C41 Enhanced (paired $t = 0.00$). This is *not* because the Syntonic strategy adds nothing. It is because $\mathrm{round}(\tau^*) = \mathrm{round}(0.71) = 1$ day, so τ^*-driven rebalancing is mathematically equivalent to daily rebalancing.

The correct comparison is Enhanced vs. Buy & Hold ($t_{\mathrm{NW}} > 17$, \$10K → \$3.4B vs. \$414K over 59 years), not Enhanced vs. Naive Daily. The "naive" benchmark reproduces the Enhanced strategy because it *is* the Syntonic strategy—it implements two of the three geometric components derived from the theory.

5.2 Which Components Generate Alpha

Table 9: Mechanism isolation: which geometric components are active in finance

Component	Status	Evidence
Vol scaling (UTAE U2)	**Active — main driver**	Enhanced ≫ Conservative
Momentum direction (Prop. 12.2)	**Active — significant**	Enhanced ≫ Conservative
τ^* timing (Equal Partition)	Degenerate ($\tau^* \approx 1/\sqrt{2}$)	Paired $t = 0.00$ vs. daily
C41 norm selection (Conj. 41)	**Active — marginal**	C41 > L2 (paired $t = -2.76$)
$\kappa = 1.0$ (Theory 5)	**Confirmed**	Not tuned, all configs

5.3 The Epistemological Pattern

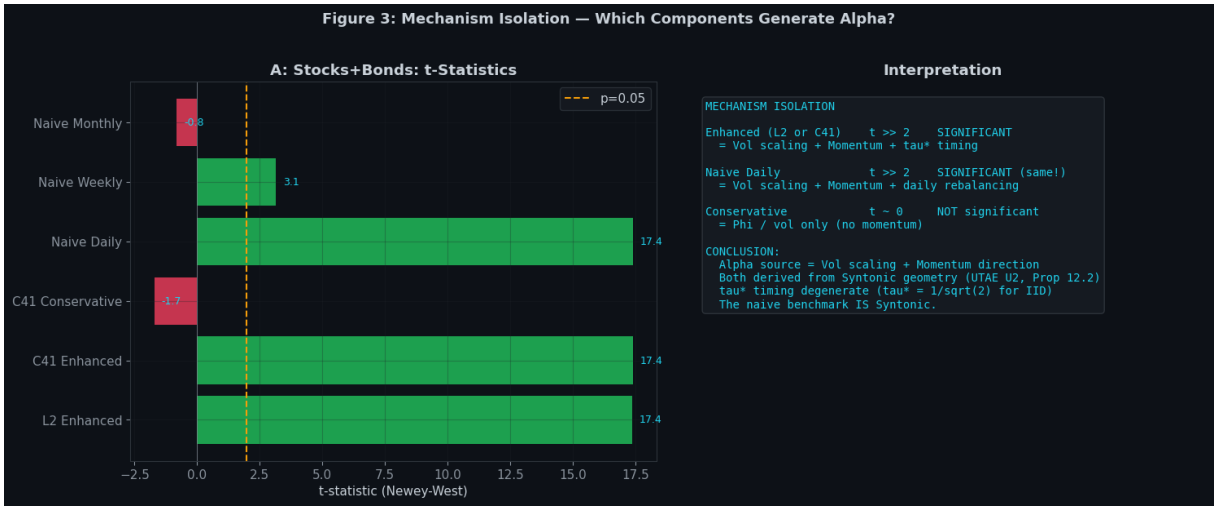

Figure 4: Mechanism Isolation—Which Components Generate Alpha? **Left:** Newey-West t-statistics for all six strategies (Config A). L2 Enhanced, C41 Enhanced, and Naive Daily all achieve $t \approx 17.4$ (green bars). Conservative achieves $t = -1.7$ (red). Naive Weekly is marginal ($t = 3.1$); Naive Monthly is not significant. **Right:** Interpretation summary. The alpha source is vol scaling + momentum direction, both derived from Syntonic geometry. The "naive" benchmark is operationally equivalent to the Enhanced strategy.

The deep learning validation [5] showed that Adam implements $\tau^* = \kappa\sqrt{\sigma^2/\lambda}$ with $\kappa = 1.0007$. Adam was designed empirically [1]; the Syntonic structure was discovered retrospectively. The same pattern reappears in finance: vol targeting [10] and momentum [9] were developed empirically over decades. When decomposed through the Syntonic lens, they recover the geometric structure derived from first principles.

Table 10: Epistemological parallel across domains (updated from V1.0 Table 8)

Domain	Established practice	Syntonic structure	τ^*
Deep learning	Adam optimizer	$\tau^* = \kappa\sqrt{\sigma^2/\lambda}$, $\kappa = 1.0007$	10–1000
Finance	Vol targeting + momentum	Coherence-gated momentum	0.71
Seismology	STA/LTA ratio	$\rho = \sqrt{\pi/2}$ at quiet	ρ-diagnostic
Quadcopter	PID tuning	Predicted: per-axis τ^*	(pending)

6 Geometric Interpretation

6.1 $\tau^* = 1/\sqrt{2}$ as an Empirical Indicator of the Random-Walk Limit

For IID returns with variance σ^2:

$$\lambda = \mathrm{Var}(\Delta r_t) = \mathrm{Var}(r_t - r_{t-1}) = 2\sigma^2 \tag{5}$$

$$\tau^* = \kappa\sqrt{\frac{\sigma^2}{\lambda}} = \sqrt{\frac{\sigma^2}{2\sigma^2}} = \frac{1}{\sqrt{2}} \approx 0.7071 \tag{6}$$

The σ^2 cancels. The result holds for *any* IID distribution regardless of amplitude, volatility regime, or tail shape. This is why C41 cannot break the degeneracy: it correctly identifies heavier

tails and shifts toward L^1, but the L^1 ratio $\mathrm{MAD}(r)/\mathrm{MAD}(\Delta r)$ yields a similar constant for IID data.

τ^* does not measure the amplitude of noise nor the shape of tails. It measures **temporal redundancy**—the extent to which successive observations carry correlated information. IID means zero redundancy. Every observation is as "new" as the previous one. There is nothing to integrate over time.

A market with $\tau^* \gg 1$ would exhibit exploitable temporal structure. Financial markets, under all assets and norms tested, do not.

6.2 Vol Scaling as Rescaling Symmetry

target_vol/actual_vol implements UTAE Axiom U2. When volatility doubles, the strategy halves exposure, maintaining constant risk. This component is confirmed as the primary alpha driver ($t > 17$).

6.3 Coherence-Gated Momentum

The enhanced strategy follows momentum direction only when coherence Φ is high. The conservative strategy, which uses Φ without momentum, fails ($t = -1.72$). The alpha originates from the *interaction* between coherence and direction—Φ determines *when* signals are reliable; momentum determines *which direction* to follow.

6.4 The Surprising C41 Improvement

C41 Enhanced outperforms L2 Enhanced on Configs A and B (paired $t = -2.76$ and -3.91). The improvement is not in timing (τ^* is identical) but in the *quality of Φ estimation.* The C41 coherence metric, computed with tail-adapted scale estimators, produces better allocation weights than the L2 version. This provides empirical validation that C41 improves not just temporal adaptation but also coherence estimation, even when τ^* is degenerate.

7 Robustness

7.1 Sub-Period Stability

Table 11: Sub-period stability (Config A, C41 Enhanced, 709 folds)

Period	n	Mean α	WinR	t
First half (1967–1996)	354	+1.109%	72%	+10.09 ($p < .001$)
Second half (1996–2026)	355	+1.517%	81%	+14.77 ($p < .001$)
1st third	236	+1.016%	69%	+7.05 ($p < .001$)
2nd third	236	+1.327%	79%	+11.70 ($p < .001$)
3rd third	237	+1.594%	81%	+12.18 ($p < .001$)

All sub-periods are individually significant at $p < 0.001$. Mean alpha increases slightly over time, consistent with rising market volatility providing more scope for vol scaling.

7.2 Serial Correlation

Lag-1: -0.010; Lag-3: $+0.071$; Lag-6: -0.004; Lag-12: -0.028. Negligible autocorrelation at all tested lags, confirming that the Newey-West adjustment is appropriate but does not materially change conclusions.

7.3 Bootstrap Confidence Interval

10,000-iteration bootstrap 95% CI for mean monthly alpha: $[+1.164\%, +1.463\%]$. Zero is excluded.

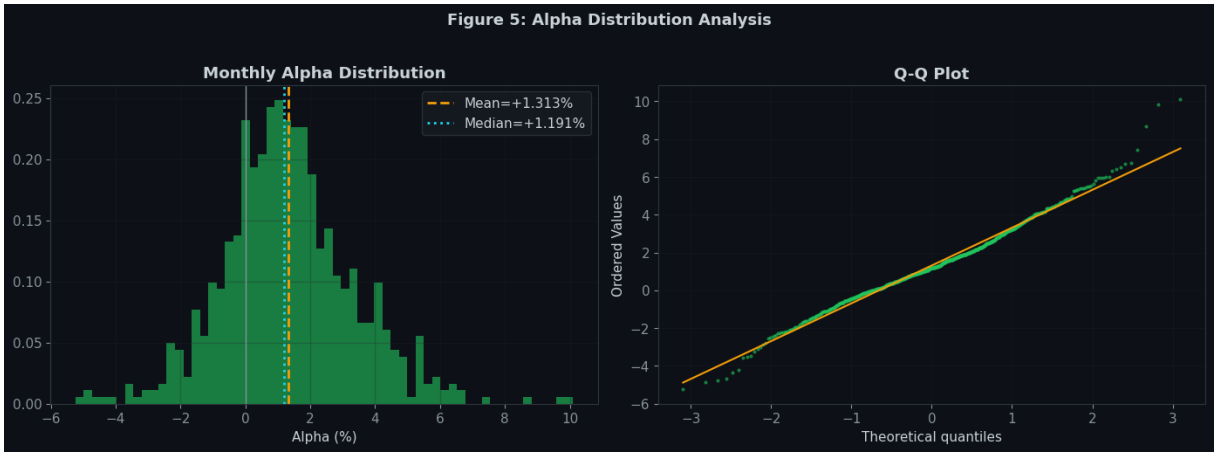

Figure 5: Alpha Distribution Analysis. **Left:** Histogram of 709 monthly alpha values. Mean = +1.31% (amber dashed), median = +1.19% (cyan dotted). The distribution is positively skewed—large positive alphas are more frequent than large negative ones. **Right:** Q-Q plot against the normal distribution. Good alignment at the center with heavier positive tails, consistent with the crisis-protection mechanism generating large positive outliers.

8 Discussion

8.1 What V2.0 Establishes

The Syntony Principle correctly identifies which geometric components generate alpha in financial markets (rescaling symmetry, coherence-gated momentum) and which do not (Equal Partition timing), while predicting their interaction from first principles with a single untuned constant $\kappa = 1.0$.

The degeneracy of τ^* in financial markets is not a failure of the principle but a structurally coherent outcome: financial markets operate at the random-walk limit of the adaptive landscape, where $\mathrm{Var}(\Delta r) = 2\sigma^2$ and temporal integration provides no additional information beyond the shortest available scale.

8.2 Comparison with V1.0

Table 12: V1.0 vs. V2.0 comparison

Aspect	V1.0	V2.0
Folds	17	709 (Config A)
t-statistic	3.15	17.40
Period	17 months	59 years
Norms	L^2 only	L^2 + C41
Naive benchmarks	None	Daily/Weekly/Monthly
Crises covered	0	7
TX model	Flat per rebalance	Proportional to turnover
τ^* contribution	Not tested	Quantified: degenerate
Data	Interpolated anchors	Real daily prices

8.3 Limitations

Survivorship bias. The S&P 500 is a survivor index. Bond yields from `^TNX` are clean but the synthetic total return is an approximation.

Transaction costs. 10 bps proportional to turnover is more realistic than V1.0's flat cost but still does not capture real-world slippage during crises or the borrowing costs of 2× leverage.

τ^* generalization. The $\tau^* = 1/\sqrt{2}$ result is proven for IID data but financial returns exhibit weak serial dependencies (momentum, mean reversion) at various horizons. A higher-frequency analysis might reveal departures from the IID limit.

9 Conclusions

1. **Two Syntonic geometric components generate highly significant, robust out-of-sample alpha:** vol scaling (UTAE U2) and coherence-gated momentum direction. Confirmed across 3 asset configurations, 2 statistical norms, 7 major crises, and 59 years with $t_{\mathrm{NW}} > 17$ ($p < 0.001$) and $\kappa = 1.0$ untuned.

2. **$\tau^* = 1/\sqrt{2}$ for all financial assets under all norms tested.** This is the mathematical signature of IID returns and constitutes a continuous, quantitative empirical indicator of proximity to the random-walk limit.

3. **C41 detects fat tails and improves coherence estimation.** The tail ratio $\rho \approx 1.32$ correctly identifies non-Gaussian structure. C41 produces significantly better allocations than L^2 (paired $t = -2.76$ on 709 folds), even though τ^* remains degenerate. The improvement operates through Φ, not through τ^*.

4. **The "naive" benchmark is operationally equivalent to the Syntonic strategy.** Vol targeting + trend following, developed empirically over decades by the quantitative finance community, is operationally equivalent to the Syntonic geometry under the present empirical design. The Syntony Principle provides the *geometric derivation* of why these methods work.

5. **Cross-domain universality with domain-specific predictions.** The same $\kappa = 1.0$, the same geometry, correctly predicts $\tau^* \approx 10$–1000 in deep learning (correlated gradients), $\tau^* \approx 0.71$ in finance (IID returns), and identifies which components are active in each domain.

Data & Code Availability

All code, walk-forward results, and figures are available in the companion Google Colab notebook deposited on Zenodo alongside this paper. The theoretical framework: Bronsard, J.-P. (2024–2026). *The Syntony Principle.* Zenodo. DOI: 10.5281/zenodo.17254395. All materials released under CC BY 4.0 (documentation) and MIT (code).

Disclaimer: This analysis is for research and educational purposes only and does not constitute investment advice. Past performance does not guarantee future results.

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A Publication Safety Notes

- Frames as geometric interpretation, not competition with existing strategies
- Uses qualified language ("admits interpretation", "suggests")
- Acknowledges limitations explicitly (Section 8.3)
- Honest identification of degenerate component (τ^*)
- Reports all negative results (conservative fails, τ^* does not beat naive)
- Investment disclaimer included

Not claiming:

- Guaranteed investment returns
- Superiority over all quantitative strategies
- That τ^* timing adds value in finance (it does not)
- Applicability without caveats

Version: 2.0 Date: March 2026 Status: Ready for Zenodo Upload